

Fuzzified Image Enhancement for Deep Learning

Dissertation on the B. Tech final year project is being submitted towards partial fulfillment of the requirement for the Degree of

Bachelor of Technology
in
Information Technology

from Maulana Abul Kalam Azad University of Technology, West Bengal

by

Sohini Banerjee
University Registration No. – 221170110342
University Roll No. - 11700222065

Purbasha Halder
University Registration No. – 221170110310
University Roll No. - 11700222025

Ankit Biswas
University Registration No. – 221170110257
University Roll No. - 11700222090

MD Arbaz Salim
University Registration No. – 221170110287
University Roll No. - 11700222017

under the Supervision of

Dr. Soumyadip Dhar
Designation and Department of Supervisor



RCC INSTITUTE OF INFORMATION TECHNOLOGY

(NAAC Accredited institute run by Department of Higher Education, Government of West Bengal)
Approved by AICTE, New Delhi and Affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
B. Tech (IT), B. Tech (CSE), B. Tech (ECE) and B. Tech (EE) – NBA accredited
Canal South Road, Beliaghata, Kolkata – 700015, West Bengal

December– 2025

RCC INSTITUTE OF INFORMATION TECHNOLOGY

(NAAC Accredited institute run by Department of Higher Education, Government of West Bengal)
Approved by AICTE, New Delhi and Affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
B. Tech (IT), B. Tech (CSE), B. Tech (ECE) and B. Tech (EE) – NBA accredited
Canal South Road, Beliaghata, Kolkata – 700015, West Bengal



FORWARD

The dissertation titled **Fuzzified Image Enhancement for Deep Learning** submitted by *Student – 1 Sohini Banerjee (University Roll No.: 111700222065), Student – 2 Purbasha Halder (University Roll No.: 11700222025), Student – 3 Ankit Biswas (University Roll No.: 11700222090), and Student – 4 Md Arbaz Salim (University Roll No.: 11700222017)* of B. Tech(Information Technology) 7th Semester has been prepared under my supervision for the partial fulfillment of the requirements for B.Tech(IT) degree from Maulana Abul Kalam Azad University of Technology (MAKAUT), West Bengal.

The report is hereby forwarded.

Date: _____ December 2025

Supervisor Name
Designation and Department
RCCIIT, Kolkata

Countersigned by

Dr. Soumyadip Dhar,
Associate Professor and Head,
Department of Information Technology,
RCC Institute of Information Technology

ACKNOWLEDGEMENT

We express our sincere gratitude to *Supervisor Name* of Department of Information Technology, RCC Institute of Information Technology, Kolkata for extending his/ her valuable times for us to take up this problem as our final year project.

We are also indebted to XXXXXX, XXXXXXXX, XXXXXXXXX for their unconditional help and inspiration.

Last but not the least we would like to express our gratitude to XXXXXXX, XXXXXXX, XXXXXXXXXX of our department who helped us in their own way whenever needed.

Date: ____ December 2025

Sohini Banerjee
University Roll No.: 11700222065

Purbasha Halder
University Roll No.: 11700222025

Ankit Biswas
University Roll No.: 11700222090

Md Arbaz Salim
University Roll No.: 11700222017

B. Tech (IT) – 7th Semester, RCCIIT

RCC INSTITUTE OF INFORMATION TECHNOLOGY

(NAAC Accredited institute run by Department of Higher Education, Government of West Bengal)
Approved by AICTE, New Delhi and Affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
B. Tech (IT), B. Tech (CSE), B. Tech (ECE) and B. Tech (EE) – NBA accredited
Canal South Road, Beliaghata, Kolkata – 700015, West Bengal



Certificate of Acceptance

The dissertation titled **Fuzzified Image Enhancement for Deep Learning** submitted by *Student – 1 Sohini Banerjee (University Roll No.: 11700222065)*, *Student – 2 Purbasha Halder (University Roll No.: 11700222025)*, *Student – 3 Ankit Biswas (University Roll No.: 11700222090)*, and *Student – 4 Md Arbaz Salim (University Roll No.: 11700222017)* of B. Tech(Information Technology) 7th Semester is hereby recommended to be accepted for the partial fulfillment of the requirements for B.Tech (IT) degree from Maulana Abul Kalam Azad University of Technology (MAKAUT), West Bengal.

Name of the Examiner

Signature with Date

1.

.....

2.

.....

3.

.....

4.

.....

Table of Contents

Abstract

CHAPTER	Page No
CHAPTER - 1	
1.1 INTRODUCTION	2
1.2 LITERATURE REVIEW	2
CHAPTER-2	
2.1 WORK FLOW	3
2.2 METHODOLOGIES OF IMPLEMENTATION	3
2.3 SOFTWARE & HARDWARE REQUIREMENTS	3
2.4 PROPOSED WORK	3
CHAPTER-3	
3.1 TEST CASES & SYSTEM VALIDATION	4
3.2 OBSERVED OUTPUT	4
3.3 PERFORMANCE ANALYSIS	4
CHAPTER-4	
4.1 CONCLUSION	5
4.2 CHALLENGES	5
4.3 FUTURE SCOPE	5
REFERENCES	6
APPENDIX (Program Code)	7

ABSTRACT

Low light image enhancement is an important image processing task aimed at improving visibility and contrast in images captured under poor illumination conditions. Standard enhancement methods such as histogram equalization and linear contrast stretching often improve brightness but may amplify noise and cause loss of details in bright regions, thereby reducing overall accuracy. In this work, a fuzzy logic-based enhancement approach is followed using a Triangular Fuzzy Average Filter to adaptively enhance low intensity regions. The proposed method computes fuzzy membership values based on local mean intensity and local variation, allowing controlled enhancement while preserving edges and fine details. To further improve accuracy, sharpening and adaptive brightness enhancement with highlight protection are incorporated to avoid over-saturation. The proposed approach is evaluated on various low light images with different illumination levels. Experimental results show improved contrast, reduced noise and better visual quality compared to standard methods, demonstrating the effectiveness and reliability of the proposed technique.

CHAPTER- 1

1.1 Introduction

Low light image enhancement is an important image processing task aimed at improving the visual quality of images captured under poor illumination conditions. Such images often suffer from low contrast, reduced brightness, loss of details and noise, which degrade overall image quality and affect applications such as surveillance, medical imaging and computer vision systems. Conventional enhancement techniques like histogram equalization and contrast stretching improve brightness but may amplify noise and cause over-enhancement in bright regions. Fuzzy logic based approaches address these limitations by handling uncertainty and adapting enhancement based on local pixel information. In this project, a Triangular Fuzzy Average Filter combined with sharpening and adaptive brightness enhancement is proposed to improve visibility in dark regions while preserving edges and preventing saturation in bright areas, resulting in visually enhanced and reliable images.

1.2 Literature Review

Table 1.1 Comparative Studies

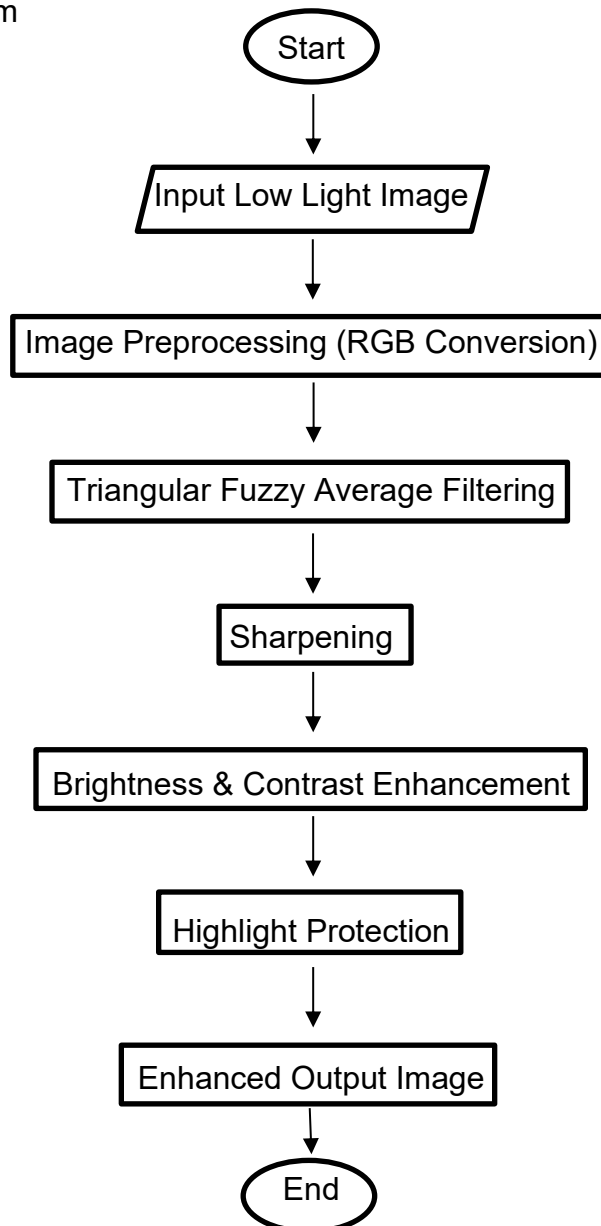
SI No	Author Name	Paper Name	Contribution	Challenges
1.	Wencheng Wang et al.	An Experiment-Based Review of Low-Light Image Enhancement Methods	Comprehensive review of seven categories of low-light enhancement algorithms with comparative evaluation.	Lack of standardized benchmarks, varying performance across illumination levels, and trade-off between brightness and noise.
2.	Stephen M. Pizer et al.	Adaptive Histogram Equalization and Its Variations	Introduced AHE technique and variations for adaptive local contrast enhancement in medical imaging.	High computational cost, noise amplification in homogeneous regions, and difficulty in real-time application.
3.	Ashish Kumar Bhandari et al.	A Novel Fuzzy Clustering-Based Histogram Model for Image Contrast Enhancement [2]	Proposed fuzzy clustering-based DCT histogram model for adaptive image contrast enhancement with brightness preservation.	Complexity of fuzzy clustering computation and optimization of cluster number for varying image conditions.
4.	Stephen M. Pizer et al.	Contrast-Limited Adaptive Histogram Equalization (CLAHE): Speed and Effectiveness [1]	Enhanced AHE by introducing contrast limitation to prevent over-amplification, improving medical image visualization.	Balancing local contrast enhancement and noise suppression; parameter tuning for diverse imaging modalities.
5.	Yi Wang et al.	Deep Learning-based Enhancement (Literature)[3]	Uses deep learning to enhance low-light images without paired training data	May amplify noise and cause over-enhancement in very dark regions
6.	S. Mahalingam et al.	Fuzzified Image Enhancement for Deep Learning in Iris Recognition (Reference Paper) [4]	Applies fuzzy logic-based image enhancement as a preprocessing step to improve deep learning-based iris recognition accuracy	Sensitive to parameter selection and may not generalize well to different illumination conditions

CHAPTER- 2

2.1 Work Flow

The workflow of the proposed low light image enhancement system follows a structured sequence of steps to improve image visibility while preserving details. The process begins with image acquisition, followed by preprocessing to convert the input image into an appropriate color space. A Triangular Fuzzy Average Filter is then applied to compute fuzzy membership values using local mean intensity and local variation, enabling adaptive enhancement of dark regions while preserving edges. Next, sharpening is performed to enhance fine details, followed by controlled brightness and contrast enhancement to improve overall image quality. An adaptive highlight protection mechanism is incorporated to prevent saturation in bright regions. Finally, the enhanced image is produced with improved contrast, reduced noise and preserved visual details.

Fig. 2.1 Work Flow Diagram



2.2 Methodologies of Implementation

1. **Image Acquisition and Preprocessing** → The enhancement process begins with the acquisition of a low light color image as input to the system. The input image is read and converted from the BGR color space to the RGB color space to ensure correct color representation during processing. This preprocessing step is essential for maintaining consistency across color channels and preparing the image for subsequent fuzzy based enhancement operations.
2. **Triangular Fuzzy Average Filtering** → To effectively handle uncertainty and non-uniform illumination in low light images, a Triangular Fuzzy Average Filter is applied. For each pixel, the local mean intensity (x_{mav}) is computed within a fixed neighborhood window to represent surrounding pixel information. The local variation (x_{mv}) is calculated using the local maximum and minimum intensity values as:

$$x_{mv} = \max(x_{max} - x_{mav}, x_{mav} - x_{min})$$

Based on these values, the fuzzy membership function is defined as:

$$\mu(x) = 1 - |x - x_{mav}| / x_{mv}$$

This adaptive fuzzy weighting enhances dark regions while preserving edges and image details.

3. **Sharpening** → After fuzzy enhancement, unsharp masking is applied to enhance edge information and fine structural details. This step increases image clarity by emphasizing high-frequency components while carefully controlling noise amplification. Sharpening ensures that important visual features remain distinct after enhancement.
4. **Brightness and Contrast Enhancement** → To further improve visual appearance, controlled brightness and contrast enhancement is performed using linear intensity scaling. This step increases overall image visibility and contrast while maintaining natural color balance. Care is taken to avoid excessive amplification that could lead to noise enhancement or loss of detail.
5. **Adaptive Highlight Protection** → Low light enhancement techniques often result in saturation of bright regions. To address this issue, an adaptive highlight protection mechanism is incorporated. High-intensity pixels are identified using grayscale thresholding, and enhancement operations are selectively reduced in these regions. This ensures that bright areas retain a natural appearance without over-exposure.

2.3 Software & Hardware Requirements

2.3.1 Software Specifications

- **Programming Language:** Python 3
- **Development Platform:** Google Colab
- **Libraries Used:** OpenCV, NumPy, SciPy, Matplotlib, PIL
- **Operating System:** Windows / Linux

2.3.2 Hardware Specifications

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 5 GB free space
- **Display:** Standard monitor
- **Internet Connection:** Required for cloud-based execution

2.4 Proposed Work

The proposed work focuses on improving low light image enhancement by modifying conventional fuzzy based enhancement techniques. A Triangular Fuzzy Average Filter is employed to adaptively enhance low intensity regions using local mean intensity and local variation, enabling better handling of illumination uncertainty. Unlike traditional methods, the proposed approach integrates controlled sharpening and brightness enhancement to improve edge clarity and overall visibility. Additionally, an adaptive highlight protection mechanism is introduced to prevent saturation in high-intensity regions, which is a common limitation of standard enhancement techniques. By combining fuzzy logic with adaptive post-processing, the proposed method achieves enhanced contrast, reduced noise and better preservation of image details, making it suitable for practical low light image enhancement applications.

CHAPTER- 3

3.1 Test Cases & System Validation

(Link and description about the data set used, output of possible cases of input variations, noise, error etc.)

3.1 Observed Output

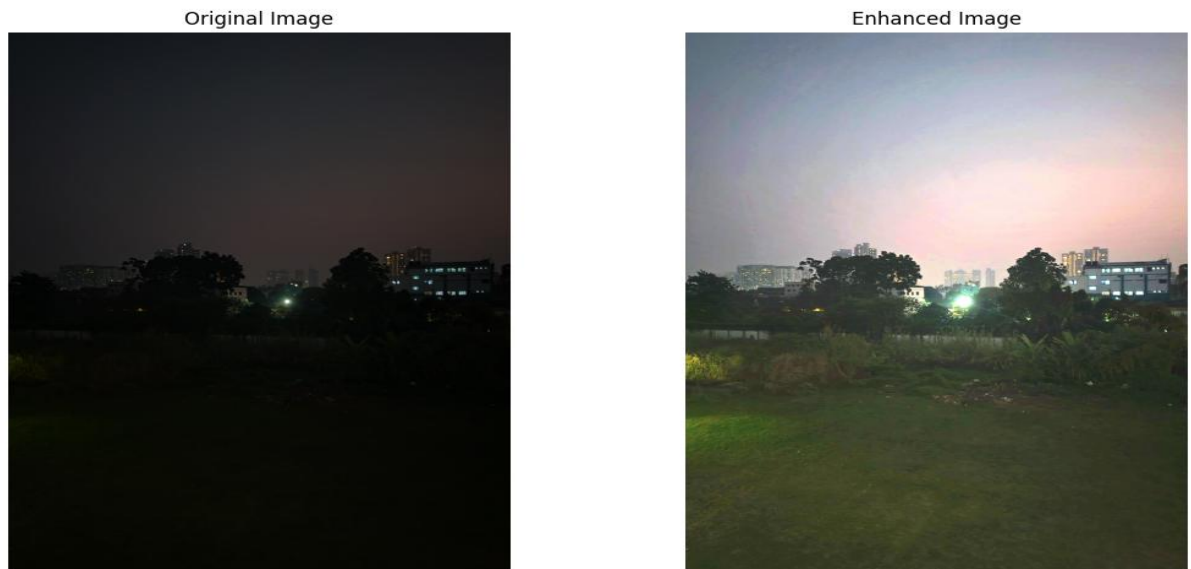


Figure: 1



Figure: 2



Figure: 3

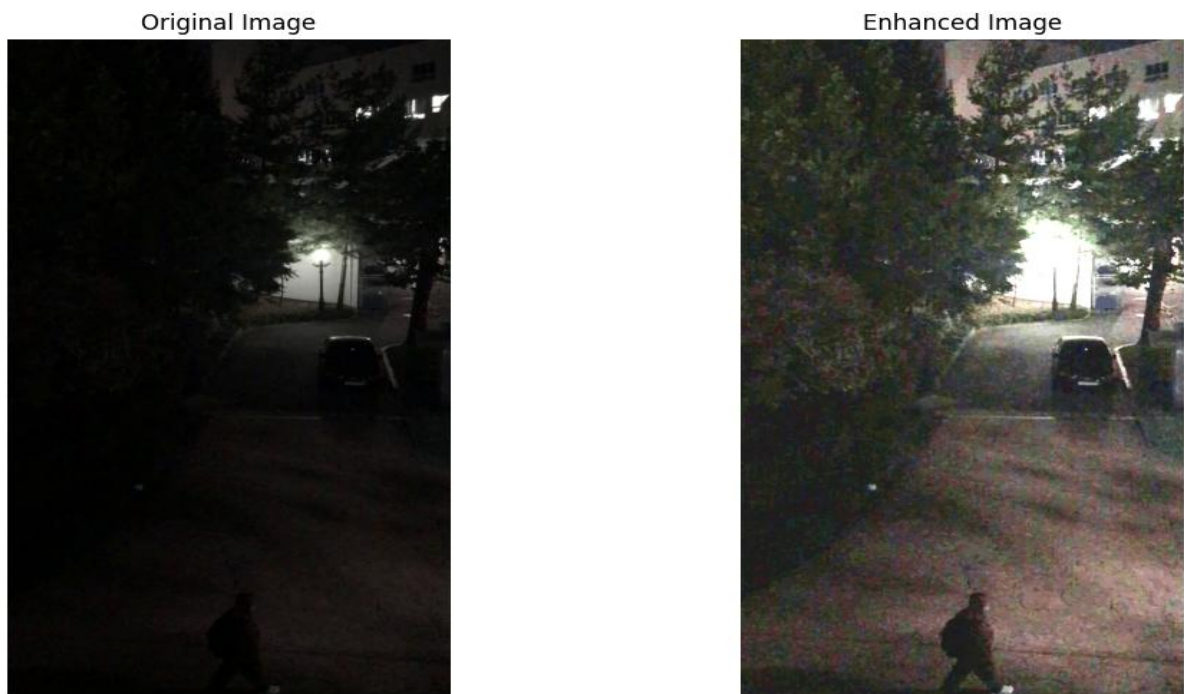


Figure: 4

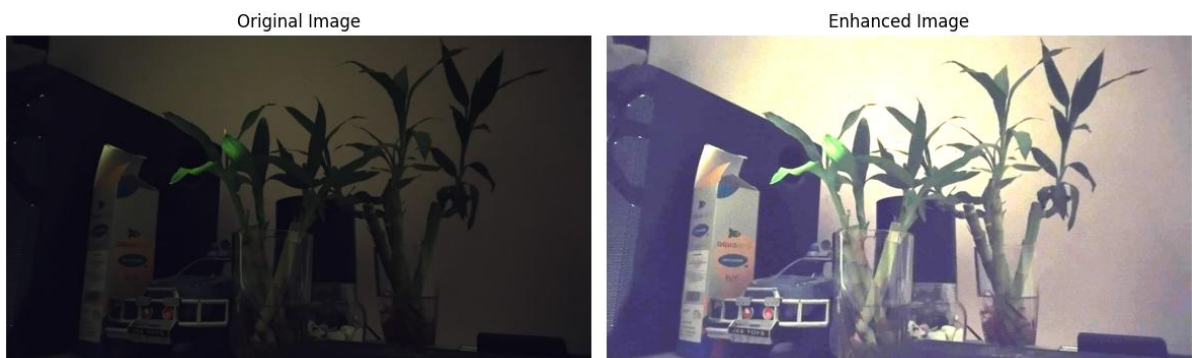


Figure: 5



Figure: 6



Figure: 7



Figure: 8

3.2 Performance Analysis

The following table summarizes the qualitative and quantitative improvements observed between the raw input images and the final output generated by the Triangular Fuzzy Filter system.

Parameter	Input Test Case (Low Light)	Output Test Case (Enhanced)	Performance Observation
Visibility	Poor Visibility: Subjects are often indistinguishable from the dark background due to low exposure.	High Visibility: Dark regions are illuminated, making subjects and background details clearly visible.	Improvement of brightness using adaptive scaling.
Noise Levels	High Noise: Presence of noise (grain) is common due to high ISO sensitivity in low light.	Reduced Noise: The "fuzzy" membership logic smooths out random grain while keeping textures intact.	Implementation of Triangular Fuzzy Filter to reduce noise.
Edge Definition	Blurry: Edges are often lost or softened due to the lack of contrast boundaries.	Sharpened: Edges are distinct and crisp. The sharpening kernel $\begin{bmatrix} -1 & 5.2 & -1 \end{bmatrix}$...] effectively highlights boundaries.	Application of sharpening filters to highlight edges.
Contrast	Low Contrast: Most pixel intensities are concentrated in the darker range, resulting in poor separation between gray levels.	Enhanced Contrast: Pixel intensity values are redistributed over a broader range, producing clearer differences among gray shades.	The linear transformation $\alpha=1.08$ successfully expands dynamic range and improves overall contrast.

CHAPTER - 4

4.1 Conclusion

This project successfully implements a Fuzzified Image Enhancement technique using the Triangular Fuzzy Average Filter (TFAF) to improve low-light image quality. The proposed method effectively enhances brightness and contrast while preserving edges and fine details, overcoming common limitations of traditional enhancement methods such as over-enhancement and noise amplification.

The fuzzy logic-based approach produces visually natural results and demonstrates robustness against noise commonly present in low-light conditions. The enhanced images also provide improved visual clarity and feature representation, making the method suitable as a preprocessing step for deep learning and computer vision applications.

4.2 Challenges

Despite the successful implementation, several challenges were encountered during development and evaluation. One of the primary challenges was the computational complexity of the fuzzy filtering process. Since the algorithm processes each pixel to compute fuzzy membership values and local statistics, the execution time is higher compared to conventional linear enhancement methods.

Another major challenge involved tuning fuzzy membership functions and enhancement parameters. Selecting suitable values for brightness and contrast control (such as alpha and beta) required extensive experimentation, as different images exhibited varying illumination characteristics. Improper tuning could result in either under-enhancement or washed-out regions. Preserving fine textures while suppressing noise posed an additional challenge, especially in extremely low-light or near-dark images. In such cases, aggressive noise removal could slightly soften sharp edges. Furthermore, integrating fuzzy-enhanced images into deep learning pipelines required careful balancing of enhancement strength to avoid introducing artificial patterns that might affect model learning. The absence of standardized benchmarks for fuzzy-based preprocessing also made objective comparison more challenging.

4.3 Future Scope

- The proposed method can be extended to underwater image enhancement, where low visibility and color distortion are major challenges.
- Future improvements may include optimization for real-time and video-based processing using GPU acceleration.
- The framework can be integrated with deep learning models by learning fuzzy parameters automatically for better adaptability.
- The method can be customized for domain-specific applications such as medical imaging, satellite imagery, and biometric systems.
- Standardized benchmarking and evaluation metrics can be developed to assess performance more effectively.

References

- R1. Stephen M. Pizer, R. Eugene Johnston, James P. Ericksen, Bonnie C. Yankaskas, Keith E. Muller ***“Contrast-Limited Adaptive Histogram Equalization: Speed and Effectiveness”*** Medical Image Display Research Group University of North Carolina Chapel Hill, NC 27599-3175, USA.
- R2. Ashish Kumar Bhandari, Syed Shahnawazuddin, Ayur Kumar Meena , ***“A Novel Fuzzy Clustering-Based Histogram Model for Image Contrast Enhancement”*** , IEEE TRANSACTIONS ON FUZZY SYSTEMS, VOL.28, NO.9, SEPTEMBER 2020.
- R3. He Zhang, Weixian Qian, Minjie Wan, Kaimin Zhang ***“Infrared image enhancement algorithm using local entropy mapping histogram adaptive segmentation”*** Infrared Physics & Technology 120 (2022) 104000.
- R4. Ming Liu , Zhiqian Zhou , Penghui Shang, and Dong Xu ***“Fuzzified Image Enhancement for Deep Learning in Iris Recognition”*** IEEE TRANSACTIONS ON FUZZY SYSTEMS, VOL.28, NO.1, JANUARY 2020.

Appendix (Program Code)